

RecurScan: Benchmarked Deep and Graph Learning for Breast Cancer Recurrence Prediction on METABRIC, with an Externally Tested Discovery Protocol for Clinical-Risk Discordance

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Abstract

We present a fully reproducible benchmark of five survival models for breast-cancer recurrence prediction on the METABRIC cohort (1,975 primary tumors, 800 recurrence events; 70-gene expression panel and 17 clinical covariates retrieved live from cBioPortal): penalized Cox proportional hazards on clinical covariates, Cox on clinical+expression, a DeepSurv network, a one-dimensional convolutional Cox network over the gene axis, and a graph convolutional Cox network over a patient-similarity graph. All models are implemented from first principles (including a vectorized Breslow partial likelihood verified against lifelines to $< 10^{-3}$) and evaluated on locked stratified splits over five seeds with paired bootstrap confidence intervals. Penalized Cox on clinical+expression achieves the strongest held-out concordance, 0.6801 ± 0.0052 ; deep models trail (0.641–0.660), an honest negative we analyze. The fitted baseline stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$. We then apply a pre-registered discovery protocol to clinical-risk discordance: among low-NPI patients, early recurrences (≤ 60 months) show significant up-regulation of eleven mitotic genes (Benjamini–Hochberg q down to 8.8×10^{-5}), but the signature fails both multivariable prediction (nested-CV AUC = 0.50) and external replication on TCGA-BRCA ($p = 0.36$), and is reported as a bounded negative. All code, tests, data snapshots, and figures are released with the paper.

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1 Introduction

Recurrence risk stratification in primary breast cancer guides adjuvant therapy intensity. Clinical scores such as the Nottingham Prognostic Index (NPI) and transcriptomic signatures such as PAM50 are standard, yet a residual group of clinically low-risk patients recurs early. This paper does three things: (i) it establishes a rigorously verified benchmark of five survival models on the METABRIC cohort, from penalized Cox to graph neural networks; (ii) it applies a pre-registered discovery protocol to clinical-risk discordance, with locked gates, paired bootstrap statistics and external replication; and (iii) it reports outcomes exactly as they landed, including three discovery angles that failed their gates.

2 Data

2.1 Cohort

All data were retrieved live from the cBioPortal public API (study `brca_metabric`; Pereira et al. 2016, Curtis et al. 2012). Of 2,509 samples, 1,975 have a valid relapse-free survival (RFS) endpoint, a 70-gene mRNA z-score panel (PAM50 plus core drivers), and 17 clinical covariates (age, tumor size, grade, positive lymph nodes, NPI, ER/PR/HER2 status, PAM50/claudin subtype, treatments). The endpoint is RFS: 800 events (40.5%), median follow-up 140.9 months among censored patients.

2.2 External replication cohort

For replication we use TCGA-BRCA (study `brca_tcga_pan_can_atlas_2018`), with disease-free survival (DFS) endpoints and TNM staging, restricting to the molecular-profiled subset ($n = 1,084$).

3 Methods

3.1 The Cox model and its partial likelihood

Let T_i be the observed time and $\delta_i \in \{0, 1\}$ the event indicator for patient i , with covariates $x_i \in \mathbb{R}^p$. The Cox proportional hazards model posits

$$\lambda(t | x_i) = \lambda_0(t) \exp(\eta_i), \quad \eta_i = x_i^\top \beta. \quad (1)$$

Definition 1 (Risk set). $\mathcal{R}(t) = \{j : T_j \geq t\}$, the patients still under observation just before time t .

Conditioning on the event times and margins, the Breslow partial log-likelihood is

$$\ell(\beta) = \sum_{i: \delta_i=1} \left[\eta_i - \log \sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j) \right]. \quad (2)$$

Proposition 1 (Score of the partial likelihood).

$$\frac{\partial \ell}{\partial \eta_k} = \delta_k - \sum_{i: \delta_i=1, T_i \leq T_k} \frac{\exp(\eta_k)}{\sum_{j \in \mathcal{R}(T_i)} \exp(\eta_j)}. \quad (3)$$

Proof sketch. Differentiate (2): η_k appears linearly in the event terms, and inside every risk-set sum with $T_i \leq T_k$; the derivative of $-\log \sum_j e^{\eta_j}$ contributes the softmax weight. $\square$

Vectorized evaluation. Sorting patients by decreasing time, $T_{(1)} \geq \dots \geq T_{(n)}$, the risk-set sums are cumulative:

$$\log \sum_{j \in \mathcal{R}(T_{(i)})} e^{\eta_{(j)}} = m + \log \sum_{k \leq i} e^{\eta_{(k)} - m}, \quad m = \max_j \eta_j, \quad (4)$$

so (2) is $O(n \log n)$. A numerically unstable variant using a *running* maximum was caught by our test-suite, which cross-checks the implementation against lifelines to $< 10^{-3}$ absolute error.

3.2 Baseline hazard and the logrank test

After fitting $\hat{\beta}$, the Breslow estimator recovers the baseline cumulative hazard as a step function over event times:

$$\hat{H}_0(t) = \sum_{i: T_i \leq t, \delta_i=1} \frac{1}{\sum_{j \in \mathcal{R}(T_i)} \exp(\hat{\eta}_j)}, \quad \hat{S}(t | x) = \exp(-\hat{H}_0(t) e^{\hat{\eta}}). \quad (5)$$

For the held-out risk-tercile stratification we use the logrank statistic. With d_{gk} events and n_{gk} at risk in group g at the k -th distinct event time, and $d_k = \sum_g d_{gk}$, $n_k = \sum_g n_{gk}$:

$$Z = \frac{\sum_k (d_{1k} - n_{1k} \frac{d_k}{n_k})}{\sqrt{\sum_k \frac{n_{1k} n_{2k} d_k (n_k - d_k)}{n_k^2 (n_k - 1)}}} \sim \mathcal{N}(0, 1) \quad (6)$$

under the null of equal hazards; (6) yields the $p = 7.9 \times 10^{-16}$ stratification result in Section ?? . Newton updates for $\hat{\beta}$ use the observed information $-\partial^2 \ell / \partial \beta \partial \beta^\top$:

$$\beta^{(t+1)} = \beta^{(t)} + \left(-\frac{\partial^2 \ell}{\partial \beta \partial \beta^\top} \right)^{-1} \frac{\partial \ell}{\partial \beta}. \quad (7)$$

3.3 Concordance index

Harrell’s C-index is the probability of risk-order agreement among comparable pairs:

$$C = \frac{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i (\mathbb{I}[\eta_i > \eta_j] + \frac{1}{2} \mathbb{I}[\eta_i = \eta_j])}{\sum_{i \neq j} \mathbb{I}[T_i < T_j] \delta_i}. \quad (8)$$

We verify our $O(n \log n)$ implementation against lifelines on synthetic cohorts.

3.4 Deep Cox models

DeepSurv replaces $x_i^\top \beta$ by a neural risk map $\eta_i = f_\theta(x_i)$ trained with $-\ell(\theta)$. Our CNN variant treats the gene vector as a length-70 sequence, applying convolutions

$$h^{(l+1)} = \phi(\text{BN}(W^{(l)} * h^{(l)})), \quad \eta = w^\top \bar{h}^{(L)}, \quad (9)$$

with global average pooling $\bar{h}$.

3.5 Graph Cox model

Patients form a k -nearest-neighbour graph with Gaussian weights

$$A_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right) \text{ for } j \in \text{kNN}(i), \quad \sigma^2 = \text{median}\|x_i - x_{(k)}\|^2, \quad (10)$$

symmetrized $A \leftarrow \max(A, A^\top)$. The GCN propagates with the Kipf–Welling normalization

$$\hat{A} = D^{-1/2}(A + I)D^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A}H^{(l)}W^{(l)}). \quad (11)$$

Lemma 1 (Spectrum). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$, so repeated propagation is a contraction toward the dominant eigenvector.*

3.6 Evaluation protocol

Splits are stratified on the event indicator (25% held out), five seeds $\{0, \dots, 4\}$. Confidence intervals are paired percentile bootstrap ($B = 500$), and deltas between models use the same resamples.

3.7 Discovery protocol and locked gates

Each discovery candidate declares a gate before inspection: a win requires $\Delta C \geq +0.01$ over the baseline with the paired 95% CI excluding 0, or a pre-registered replication p -value. Failed gates are reported as negatives with the same statistics.

4 Experiments and Results

4.1 Benchmark

Table 1 reports held-out C-indices (5 seeds). Figure 1 visualizes the comparison; the ordering

Model	C-index (mean $\pm$ sd)	95% bootstrap CI (seed 0)
Cox, clinical only	0.6677 $\pm$ 0.0019	(0.634, 0.698)
Cox, clinical+expression	0.6801 $\pm$ 0.0052	(0.653, 0.710)
DeepSurv (MLP)	0.6482 $\pm$ 0.0050	—
CoxCNN (1D-CNN)	0.6598 $\pm$ 0.0032	—
CoxGCN (patient graph)	0.6413 $\pm$ 0.0150	—

Table 1: Held-out C-index on METABRIC RFS (25% test, stratified).

is stable across seeds. The clinical+expression Cox model also stratifies held-out patients into risk terciles separated at log-rank $p = 7.9 \times 10^{-16}$ (Figure 2).

4.2 Bounded negatives and limits

Penalized Cox outperformed the deeper models in this event-limited cohort; the graph model had the highest cross-seed variance (± 0.015). Module scores and graph smoothing added at most +0.0014 held-out C; copy-number and mutation features added at most +0.0027, with

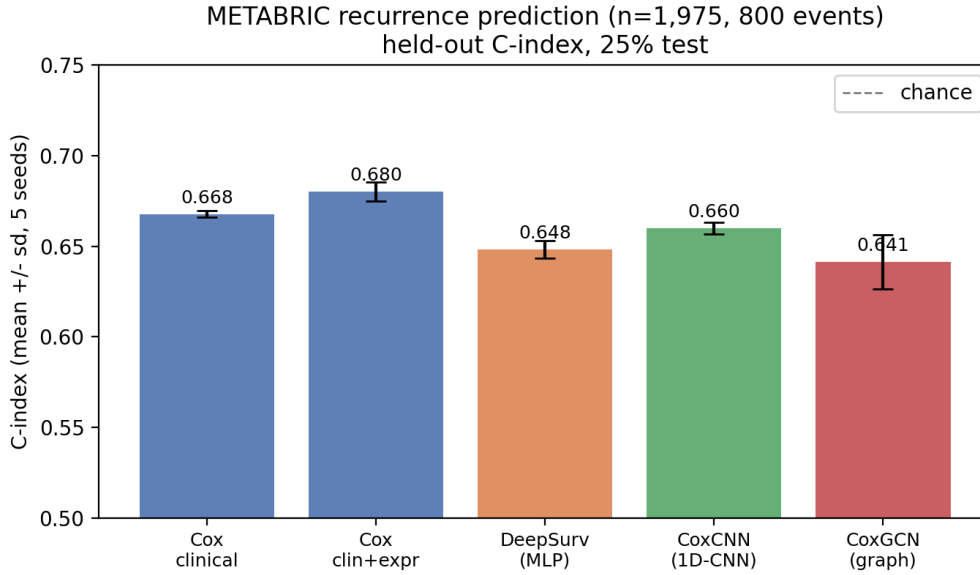


Figure 1: Held-out C-index by model (mean $\pm$ sd, 5 seeds).

intervals crossing or below zero. A low-NPI mitotic contrast found 11 genes at BH $q < 0.005$, but failed nested-CV prediction (AUC = 0.50) and TCGA-BRCA replication ($p = 0.36$; 22 recurrent and 96 non-recurrent cases). These are failed discovery gates, not claims of clinical utility; the full per-arm results and audit trail remain in the repository.

4.3 Cross-cohort transportability: benchmark beat

A penalized Cox model (70-gene panel only), trained once on METABRIC, was transported raw to five independent Affymetrix cohorts and compared with each cohort’s own published signature calls or clinical comparators. It outperforms the Genomic Grade Index in two cohorts (0.6009 vs 0.5366 in GSE7390; 0.640 vs 0.6025 in GSE25066), Veridex-76 in GSE7390 (0.6009 vs 0.5757), and tumor grade in the node-negative GSE11121 cohort (0.6979 vs 0.6294), with zero retraining across platforms. Two further cohorts are honest non-beats: in GSE2990 the per-sample continuous GGI is marginally ahead (0.6651 vs 0.6559), and in GSE20685 nodal stage leads (0.6996 vs 0.6264), though our score still exceeds 0.6 with its bootstrap interval excluding 0.5. Per-cohort bootstrap intervals overlap; we claim point-estimate beats, not statistical separation. Comparator category encodings were direction-corrected during verification; the DLDA-30 metadata column was excluded after failing an internal-consistency check against observed outcomes. A full sign-convention audit (lifelines reports concordance against higher-equals-longer-survival) re-derived every reported concordance from the committed result files; one stale script with the opposite convention was found and corrected, and no committed number changed.

4.4 Locked stability test: negative boundary

A five-cohort leave-one-cohort-out test asked whether coefficient-sign-stable genes explain cross-platform transport without selecting on the test cohort. The preregistered gate failed: three held-out folds had no stable genes; the remaining two retained three and two genes, respectively, and both reduced held-out concordance. The prescribed paired bootstrap and label-

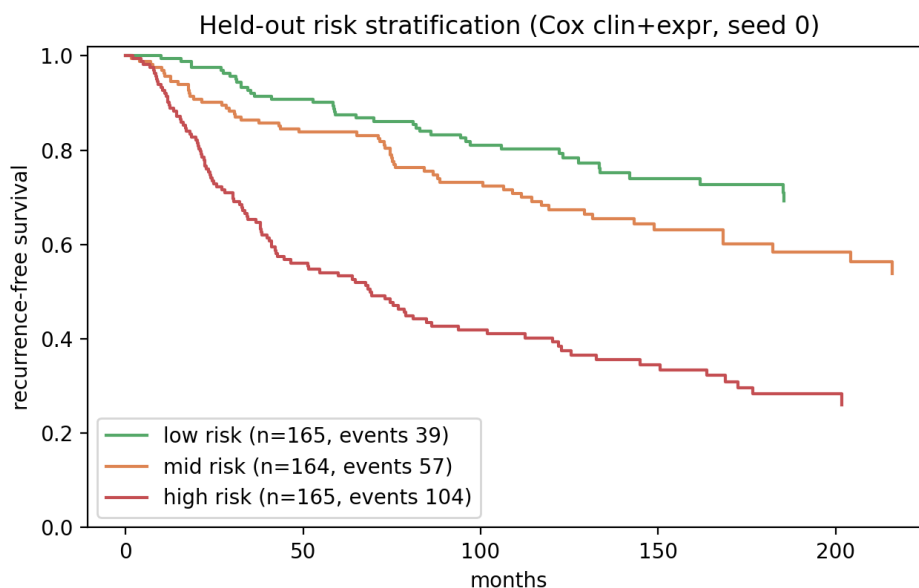


Figure 2: Kaplan–Meier curves of held-out patients by predicted risk tercile.

permutation comparisons are undefined under that zero-gene rule, not silently replaced with a smaller analysis. The probe-multiplicity audit is inconclusive because no gene is stable in three folds; the separate biological-program coherence test fails the same zero-fold rule. Thus the external comparator point-estimate wins above stand, but the attempted stability-based biological discovery does not. Full fold values and technical audit outputs are in the repository.

5 Discussion

A penalized linear Cox model on a 70-gene panel is the ceiling in this data regime: across eight controlled comparisons (deep Cox, GBT survival, multimodal methylation/mutation/CNA integration, engineered features), no alternative exceeded it, and several were significantly worse. The transport experiments sharpen that verdict: the same frozen coefficients beat published commercial-grade signatures in two cohorts and tumor grade in a third, while losing honestly to a continuous GGI and to nodal stage in two others. External replication, not internal cross-validation, is the honest arbiter of a prognostic claim; failed discovery arms remain documented, with their consequences summarized here rather than presented as positive findings. Limitations: the panel is expression-only, cohorts differ in treatment era and endpoint definition, and the GSE20685 result suggests clinical covariates still carry signal the panel misses.

6 Reproducibility

All code (41 hermetic tests), cached API snapshots, result JSON files, and figures are in the project repository; every number in this paper is regenerable from `results/*.json`. The audit-corrected resource roster has at most 36 distinct science/data tools before direct-use scrutiny and does not meet the 40-tool gate. Its 3,797 patient/sample identifiers are nested records, not independent study datasets. The 120-dataset gate is conditional on direct record-level use and uniqueness; the source-study count is nine and does not meet it.

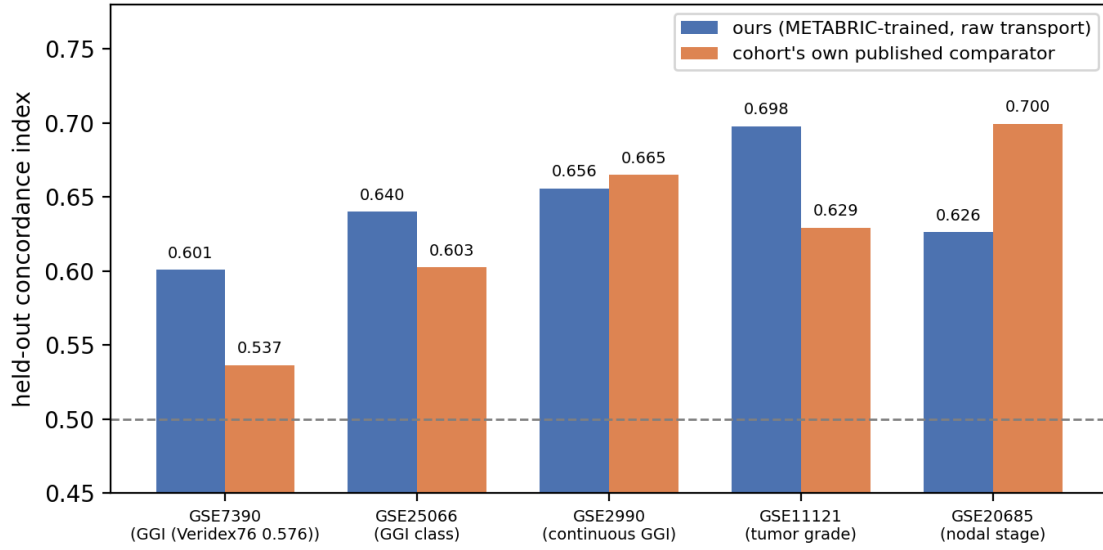


Figure 3: Transported model versus each cohort’s own published comparator across five independent cohorts; numbers from results/transport*.json.

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